

- Keep the elbows pointed out to the sides as the bar brushes against the body.
- Keep the torso and knees in the same position.
- Do not rise up on the toes or swing the bar upward.
- At the highest bar position, the elbows should be level with or slightly higher than the shoulders and wrists.

Downward Movement Phase

- Allow the bar to slowly descend back to the starting position.
- Keep the torso and knees in the same position.

MAJOR MUSCLES INVOLVED

deltoids, upper trapezius

VERBAL CUES

- **Lead with your elbows**
- **Keep the bar close to your body**
- **Think about drawing a straight line up your torso with the bar**

COMMON ERRORS

- **Not maintaining an erect torso throughout the movement**
- **Not keeping the bar close to the body**
- **Lifting through the wrists and not the elbows**



Starting position



Upward and downward movements

37

LATERAL SHOULDER RAISE

Starting Position

- Grasp two dumbbells with a closed, neutral grip.
- Position the feet shoulder- or hip-width apart, with the knees slightly flexed, torso erect, shoulders back, and eyes focused ahead.
- Move the dumbbells to the front of the thighs, positioning them with the palms facing each other.
- Slightly flex the elbows and hold this flexed position throughout the exercise. (*Note:* The elbows should be slightly more flexed than what is seen in the photos.)

Upward Movement Phase

- Raise the dumbbells up and out to the sides; the elbows and upper arms should rise together ahead of the forearms, hands, and dumbbells.
- Maintain an erect upper body position with the knees slightly flexed and feet flat.
- Do not jerk the body or swing the dumbbells upward.
- Continue raising the dumbbells until the arms are approximately parallel to the floor or nearly level with the shoulders.

Downward Movement Phase

- Allow the dumbbells to descend slowly back to the starting position.
- Keep the torso and knees in the same position.

MAJOR MUSCLES INVOLVED

deltoids

VERBAL CUES

- **Raise your arms to shoulder height**
- **Imagine pouring water from pitchers**
- **Lower slowly**

COMMON ERRORS

- **Swinging the body to lift the weights instead of using controlled shoulder movement**
- **Raising the arms above shoulder level**
- **Dropping the weights quickly instead of lowering them under control**



Starting position



Upward and downward movements

Triceps

38 LYING BARBELL TRICEPS EXTENSION

Starting Position: Athlete

- Lie in a supine position on a bench in the five-point body contact position.
- Grasp the bar from the spotter with a closed, pronated grip about 12 inches (30 cm) wide.
- Position the bar over the chest with the elbows fully extended and the arms parallel.
- Point the elbows toward the knees (not out to the sides).
- All repetitions begin from this position.

Starting Position: Spotter

- Stand erect and very close to the head of the bench (but do not distract the athlete).

- Place the feet shoulder-width apart in a staggered stance with the knees slightly flexed.
- Grasp the bar with a closed, alternated grip.
- Hand the bar to the athlete.
- Guide the bar to a position over the athlete's chest.
- Release the bar smoothly.

Downward Movement Phase: Athlete

- Keeping the upper arms stationary, allow the elbows to slowly flex to lower the bar toward the face.
- Keep the wrists stiff and the upper arms perpendicular to the floor and parallel to each other.
- Lower the bar until it almost touches the head or face.
- Maintain the five-point body contact position.

Downward Movement Phase: Spotter

- Place the hands close to—but not touching—the bar as it descends.
- Slightly flex the knees, hips, and torso and keep the back neutral when following the bar.

Upward Movement Phase: Athlete

- Push the bar upward by extending the elbows back to the starting position.
- Keep the wrists stiff and the elbows pointed toward the knees.
- Keep the upper arms parallel to each other and perpendicular to the floor.
- Maintain the five-point body contact position.
- At the end of the set, signal the spotter to take the bar.
- Keep a grip on the bar until the spotter removes it.

Upward Movement Phase: Spotter

- Keep the hands close to—but not touching—the bar as it ascends.

- Slightly extend the knees, hips, and torso and keep the back neutral when following the bar.
- At the athlete's signal after the set is completed, grasp the bar with an alternated grip, take it from the athlete, and set it on the floor.

MAJOR MUSCLE INVOLVED

triceps brachii

VERBAL CUES

- **Elbows point toward the ceiling**
- **Push knuckles to ceiling**
- **Extend arms fully at the top**

COMMON ERRORS

- **Allowing the elbows to flare to the sides during the movement**
- **Flexing the wrists instead of keeping them in a neutral position aligned with the forearm**



Starting positions



Downward and upward movements

▶ **39** **TRICEPS PUSHDOWN (MACHINE)**

Starting Position

- Grasp the bar with a closed, pronated grip 6 to 12 inches (16-30 cm) wide.

- Stand erect with feet shoulder-width apart, knees slightly flexed. Place the body close enough to the machine to allow the cable to hang straight down when it is held in the starting position.
- Pull the bar down to position the upper arms against the sides of the torso.
- Flex the elbows to position the forearms parallel to the floor or slightly above.
- All repetitions begin from this position.

Downward Movement Phase

- Push the bar down until the elbows are fully extended.
- Keep the torso erect and the upper arms stationary.
- Do not forcefully lock out the elbows.

Upward Movement Phase

- Allow the elbows to slowly flex back to the starting position.
- Keep the torso, arms, and knees in the same position.
- At the end of the set, return the bar to its resting position.

MAJOR MUSCLES INVOLVED

triceps brachii

VERBAL CUES

- **Elbows tight to your sides**
- **Full extension at the bottom**
- **Slow on the way up**

COMMON ERRORS

- **Not fully extending the arms at the bottom of the ROM or not returning to the starting position at the top**
- **Allowing the weight to move quickly in an uncontrolled manner during the upward phase of the movement instead of controlling it**



Starting position



Downward and upward movements

Power Exercises



40

PUSH PRESS

This exercise consists of quickly and forcefully pushing the bar from the shoulders to over the head. Although the ascent consists of two phases, the upward movement of the bar occurs in one continuous motion without interruption. Both the push press and the push jerk exercises involve a rapid hip and knee extension that accelerates the bar off the shoulders, followed immediately by movements that position the bar overhead. The technique used to attain this final bar position varies, however. In the push press, the hip and knee extension thrust is only forceful enough to drive the bar one-half to two-thirds the distance overhead. From this height, the bar is pressed out to the overhead position, with the hips and knees remaining fully extended after the thrust.

For either exercise, the athlete can begin with a bar taken from shoulder-height supports outside a power rack or by lifting a bar from the floor to the shoulders (via a repetition of the power clean

exercise). Both exercises may be performed behind the neck using either a clean grip or a snatch grip. The following checklists describe the exercises using a power rack (although the rack is not shown).

Starting Position

- Grasp the bar with a closed, pronated grip.
- Grip should be slightly wider than shoulder-width (clean grip).
- Step under the bar and position the feet hip-width apart and parallel to each other.
- Move up to the bar to place it on top of the anterior deltoids and clavicles.
- Extend the hips and knees to lift the bar off the supports.
- Step back away from the supports and stand in the middle of the lifting platform.
- Position the feet hip- to shoulder-width apart and even with each other, with the toes pointed slightly outward.
- All repetitions begin from this position.

Preparation Phase: Dip

- Flex the hips and knees at a controlled speed to move the bar in a straight path downward.
- Continue the dip to a depth not to exceed a quarter squat, the catch position of the power clean, or 10% of the athlete's height.
- Keep the feet flat on the floor, the torso erect, and the elbows underneath or slightly ahead of the bar.

Upward Movement Phase: Drive

- Immediately on reaching the lowest position of the dip, reverse the movement by forcefully and quickly extending the hips, knees, and ankles and then the elbows to move the bar overhead.



Starting position



Dip

Catch (for the Push Press)

- After the hips and knees are fully extended and the bar is overhead from the drive phase, press it up the rest of the way until the elbows are fully extended.
- In this position the torso is erect, the head is in a neutral position, the feet are flat on the floor, and the bar is slightly over or behind the ears.

Downward Movement Phase

- Lower the bar by gradually reducing the muscular tension of the arms to allow a controlled descent of the bar to the shoulders.
- Simultaneously flex the hips and knees to cushion the impact of the bar on the shoulders.
- At the end of the set, step toward the rack and place the bar in the supports.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus

medialis, rectus femoris, soleus, gastrocnemius, deltoids, trapezius

VERBAL CUES

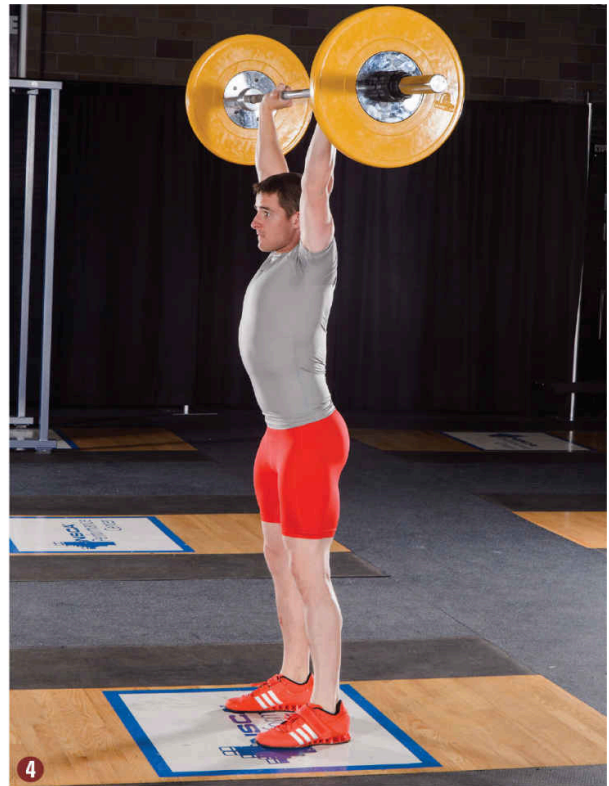
- Initiate dip with hips, not knees
- Drive from your heels
- Punch hands to sky

COMMON ERRORS

- Pushing hips back during the dip
- Dropping the elbows too much during the dip (as seen in the photos on the previous page)
- Shifting to the toes during the dip



Drive



Catch (for the push press)

Note: The upper arms should be pointing away from the body, not toward the floor, in the starting position through the dip. The result is about 45 degrees of shoulder flexion

| in the sagittal plane.



41

PUSH JERK

The push jerk involves a more forceful hip and knee thrust so that the bar is actually driven upward and caught with extended elbows in the overhead position with the hips and knees slightly flexed.

Starting Position

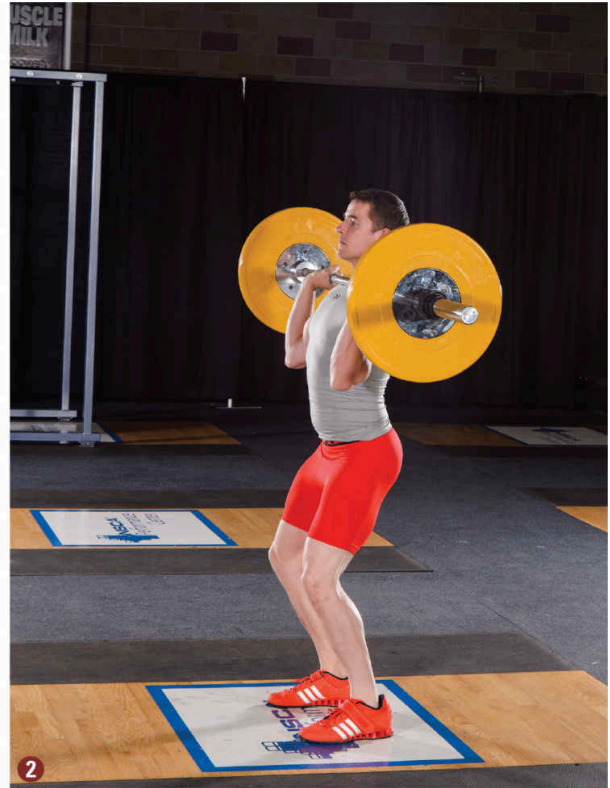
- Grasp the bar with a closed, pronated grip.
- Grip should be slightly wider than shoulder-width (clean grip).
- Step under the bar and position the feet hip- to shoulder-width apart and parallel to each other.
- Move close to the bar so that it is resting on the anterior deltoids and clavicles.
- Extend the hips and knees to lift the bar out of the supports.
- Step back away from the supports and stand in the middle of the lifting platform.
- Position the feet hip-width apart or slightly wider with toes pointed straight ahead or slightly outward.
- All repetitions begin from this position.

Preparation Phase: Dip

- Flex the hips and knees at a controlled speed to move the bar in a straight path downward.
- Continue the dip to a depth not to exceed a quarter squat.
- Keep the feet flat on the floor, the torso erect, and the elbows underneath or slightly ahead of the bar.



Starting position



Dip

Note: The upper arms should be pointing away from the body, not toward the floor, in the starting position through the dip. The result is about 45 degrees of shoulder flexion in the sagittal plane.

Upward Movement Phase: Drive

- Immediately on reaching the lowest position of the dip, reverse the movement by forcefully and quickly extending the hips, knees, and ankles, and then the elbows to move the bar overhead.

Catch (for the Push Jerk)

- After the hips and knees are fully extended and the bar is being driven overhead, quickly flex the hips and knees to a dipped position while simultaneously fully extending the elbows so that the bar is received overhead at the same moment that the bar reaches its highest position.

- Catch the bar with the torso erect, the head in neutral position, feet flat on the floor, and the bar slightly behind the head, forming a straight line through the body.

Recovery Phase

- After gaining control and balance, stand up by extending the hips and knees to a fully erect position with the feet flat on the floor.
- Keep the elbows locked while the bar is stabilized overhead.

Downward Movement Phase

- Lower the bar by gradually reducing the muscular tension of the arms to allow a controlled descent of the bar to the shoulders.
- Simultaneously flex the hips and knees to cushion the impact of the bar on the shoulders.
- At the end of the set, step toward the rack and place the bar in the supports.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris, soleus, gastrocnemius, deltoids, trapezius

VERBAL CUES

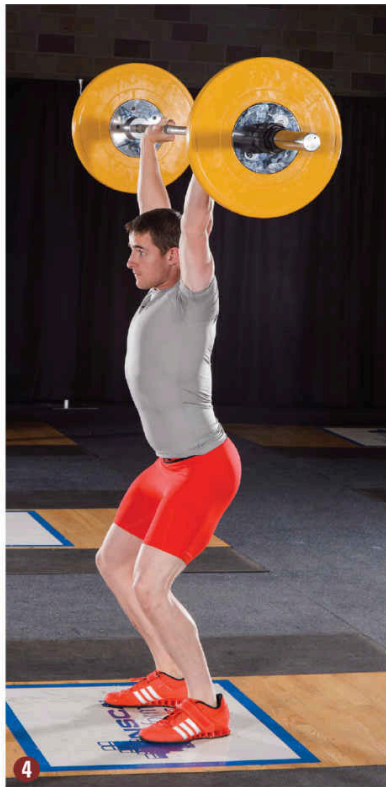
- **Initiate the dip with the hips**
- **Drive from your heels**
- **Punch your hands to the sky**

COMMON ERRORS

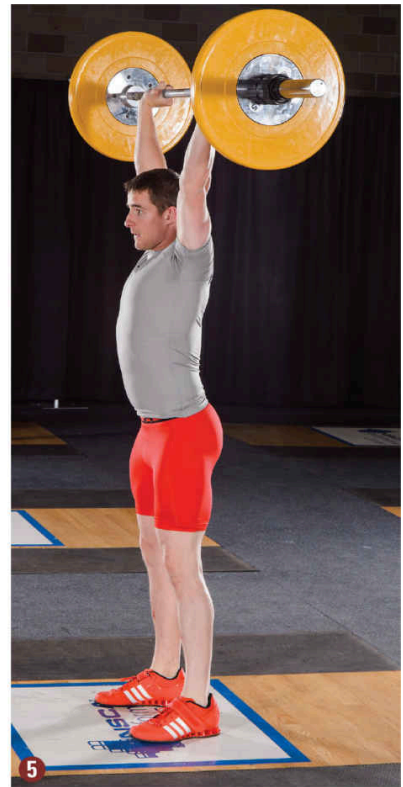
- **Pushing hips back during the dip**
- **Dropping the elbows too much during the dip (as seen in the photos on the previous page)**
- **Shifting to the toes during the dip**



3 Drive



4 Catch (for the push jerk)



5 Recovery position

▶ **42 POWER CLEAN (AND HANG POWER CLEAN VARIATION)**

This exercise consists of quickly and forcefully pulling the bar from the floor to a catching position on the front of the shoulders—all in one movement. Although the lift consists of four phases (i.e., first pull, transition, second pull, and catch) the upward movement of the bar occurs in one continuous motion without interruption. While the power clean is commonly performed by taking the barbell from the floor, it can also be performed from a hang position (i.e., at the power position or a position above or below the knee). When a hang power clean is performed, the bar is not returned to the floor or blocks between repetitions.

Starting Position

- Stand with the feet placed between hip- and shoulder-width apart with the toes pointed slightly outward (based on comfort).

- Squat down with the hips lower than the shoulders and grasp the bar evenly with a pronated grip and the shoulder directly above or slightly in front of the barbell. If a stronger grip is needed, use a hook grip.
- Place the hands on the bar slightly wider than shoulder-width apart (this can vary depending on elbow and wrist mobility and the front rack—catch position), outside of the knees, with the elbows fully extended and pointing out to the side (i.e., arm is internally rotated and the shoulder is retracted).
- Place the feet flat on the floor and position the bar approximately 1 inch (3 cm) in front of the shins and over the balls of the feet. The athlete's center of pressure on the floor should be around the bottom shoelace, toward the front of the mid-foot.
- Position the body with the
 - back neutral or slightly arched,
 - scapulae depressed and retracted,
 - chest held up and out, strongly braced,
 - head in line with the vertebral column or slightly hyperextended,
 - shoulders over or slightly in front of the bar, and
 - eyes focused straight ahead or slightly upward.
- All repetitions begin from this position.

Upward Movement Phase: First Pull

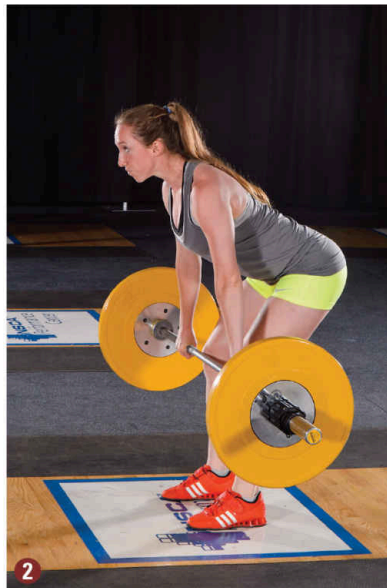
- Lift the bar off the floor by forcefully extending the hips and knees, pushing into the floor
- Keep the torso-to-floor angle constant; do not let the hips rise before the shoulders.
- Maintain a neutral spine position.
- Keep the elbows fully extended, pointing out to the side, and the shoulders over or slightly ahead of the bar.
- As the bar rises, keep it as close to the shins as possible.

- The first pull ends when knees are maximally extended, and the shins should be oriented vertically (or near vertical).

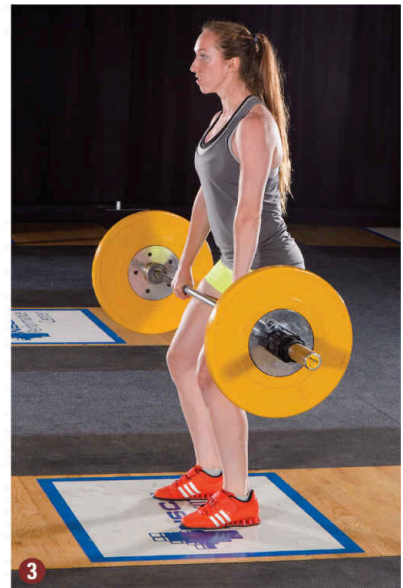
Note: A common issue beginners can struggle with is allowing the bar to pull the athlete's center of pressure on the floor forward resulting in the bar's trajectory moving away from the shin as the athlete moves through the first pull. If this occurs, the athlete will have to jump forward to catch the bar.



1 Starting position or beginning of first pull



2 End of first pull or beginning of transition



3 End of transition or beginning of second pull

Upward Movement Phase: Transition

- As the bar rises just above the knees, extend the hips, while moving the knees forward and under the bar, to transition the athlete into the power position.
- Maintain bar contact with the thighs.
- Keep the back neutral or slightly arched and the elbows fully extended and pointing to the sides.

Note: The transition phase is similar to the RDL; in fact, weightlifters use the RDL to strengthen this movement pattern.

Upward Movement Phase: Second Pull

- Rapidly extend the hips, knees, and ankles culminating with a shrug. (*Note:* It is important that the heels stay in contact with the floor for as long as possible in order to maximize force transference to the barbell.)
- Keep the bar as close to the body as possible.
- Keep the back neutral and the elbows pointing out to the sides.
- Keep the shoulders over the bar and the elbows extended as long as possible.
- When the lower body joints reach full extension, rapidly shrug the shoulders upward with the elbows still fully extended and pointing to the sides.
- As the shoulders reach their highest elevation, flex the elbows to begin pulling the body under the bar.
- Due to the explosive nature of this phase, the torso is erect or slightly hyperextended, the head may tilt slightly back, and the heels may lose contact with the floor.

Upward Movement Phase: Catch

- After the hip and knee have fully extended, pull the body under the bar and rotate the arms around and under the bar.
- Simultaneously, flex the hips and knees to a quarter-squat position.
- Once the arms are under the bar, lift the elbows to position the upper arms parallel to the floor.
- Rack the bar across the front of the clavicles and anterior deltoids.
- Catch the bar with a nearly erect torso, the shoulders slightly ahead of the hips, a neutral head position, and flat feet.
- After gaining control and balance, stand up by extending the hips and knees to a fully erect position.

Downward Movement Phase

- Lower the elbows to unrack the bar from the anterior deltoids and clavicles, then slowly lower the bar down to the thighs.

- Simultaneously flex the hips and knees to cushion the impact of the bar on the thighs.
- Squat down with the elbows fully extended until the bar touches the floor or drop the bar to the platform if rubber bumper plates are being used.

MAJOR MUSCLES INVOLVED

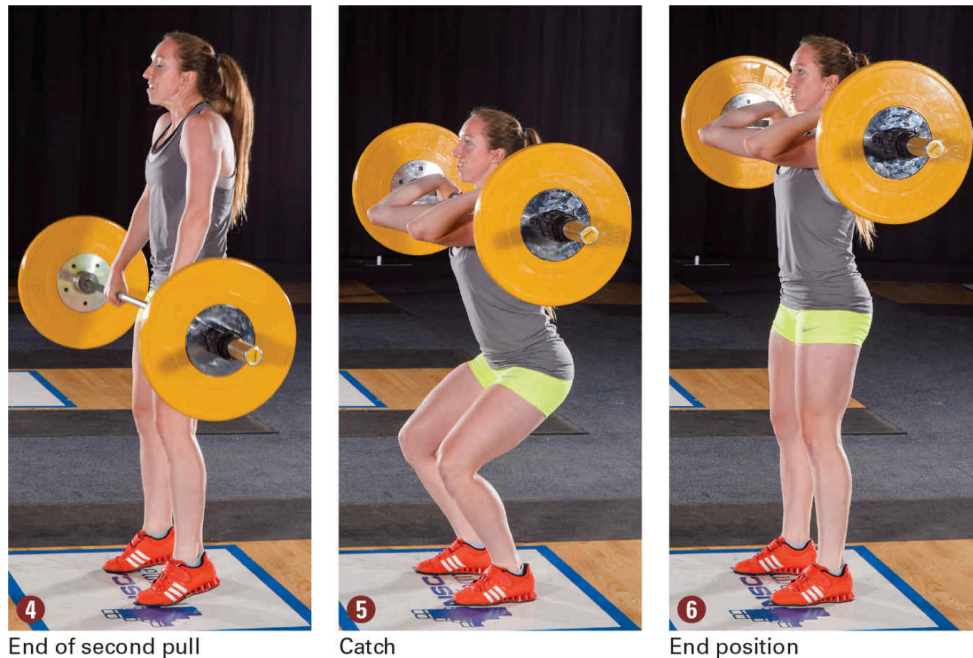
gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris, soleus, gastrocnemius, deltoids, trapezius

VERBAL CUES

- **Squeeze the bar from the floor**
- **Keep the arms long**
- **Shrug vertically**

COMMON ERRORS

- **Letting the barbell drift away from the body during the first pull**
- **Dropping the hips during the second pull**
- **Flexing the elbows before completing the triple extension**



▶ **43** **POWER SNATCH (AND HANG POWER SNATCH VARIATION)**

This exercise consists of quickly and forcefully pulling the bar from the floor to a position overhead with the elbows fully extended. Although the ascent consists of four phases (i.e., first pull, transition, second pull, and catch), the upward movement of the bar occurs in one continuous motion. While the power snatch can be initiated from the floor, it can also be initiated from a hang position (i.e., power position or slightly above or below the knee). When a hang power snatch is performed it is important to note that the bar is not returned to the floor or rack between repetitions. Regardless of where the power clean is initiated from, the bar is received in a position where the top of the thighs are above parallel.

Starting Position

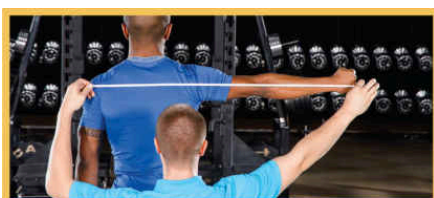
- Stand with the feet placed between hip- and shoulder-width apart with the toes pointed slightly outward.
- Squat down with the hips lower than the shoulders and grasp the bar evenly with a pronated grip. If a stronger grip is

needed, use a hook grip.

- The grip width is wider than for other exercises; a way to estimate it is to measure and use one of these distances for spacing the hands: the distance from the edge of the clenched fist of one hand to the opposite shoulder when the arm is straight out at the side, or the elbow-to-elbow distance when the arms are straight out at the side.
- Extend the elbows fully and point them out to the side (i.e., arm is internally rotated, and the shoulder is retracted).
- Place the feet flat on the floor and position the bar approximately 1 inch (3 cm) in front of the shins and over the balls of the feet.
- Position the body with the
 - back neutral or slightly arched,
 - scapulae depressed and retracted,
 - chest held up and out,
 - head in line with the vertebral column or slightly hyperextended,
 - feet flat on the floor,
 - shoulders over or slightly in front of the bar, and
 - eyes focused straight ahead or slightly upward.
- All repetitions begin from this position.

Upward Movement Phase: First Pull

- Lift the bar off the floor by forcefully extending the hips and knees.
- Keep the torso-to-floor angle constant; do not let the hips rise before the shoulders.
- Maintain the neutral spine position.
- Keep the elbows fully extended, pointing out to the side, and the shoulders over or slightly ahead of the bar.
- As the bar is raised, keep it as close to the shins as possible.



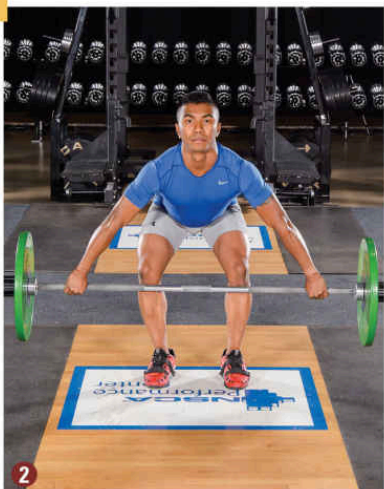
Grip measurement: fist-to-opposite-shoulder method



Grip measurement: elbow-to-elbow method



1 Starting position or beginning of first pull



2 End of first pull or beginning of transition



3 End of transition or beginning of second pull

Upward Movement Phase: Transition

- As the bar rises just above the knees, extend the hips while moving the knees forward and under the bar (the “power position”).
- Maintain bar contact with the thighs.
- Keep the back neutral or slightly arched and the elbows fully extended and pointing to the sides.

Note: The transition phase is similar to the RDL.

Upward Movement Phase: Second Pull

- Rapidly extend the hips, knees, and ankles moving into an aggressive shrug.

- Keep the bar close to the body, the back neutral, and the elbows pointing out to the sides.
- Keep the shoulders over the bar and the elbows extended as long as possible.
- When the lower body joints reach full extension, rapidly shrug the shoulders with the elbows still fully extended and pointing to the sides.
- As the shoulders reach their highest elevation, flex the elbows to begin pulling the body under the bar.
- Due to the explosive nature of this phase, the torso is erect or slightly hyperextended, the head may tilt slightly back, and the heels may lose contact with the floor.

Upward Movement Phase: Catch

- After the lower body has fully extended, pull the body under the bar and rotate the hands around and under the bar.
- Simultaneously, flex the hips and knees to a quarter-squat position.
- Once the body is under the bar, catch the bar over and slightly behind the ears with fully extended elbows, an erect and stable torso, a neutral head position, flat feet, and the body's weight over the middle of the feet.
- After gaining control and balance, stand up by extending the hips and knees to a fully erect position and stabilize the bar overhead.

Downward Movement Phase

- Lower the bar from the overhead position by gradually reducing the muscular tension of the shoulders to allow a controlled descent of the bar to the thighs.
- Simultaneously flex the hips and knees to cushion the impact of the bar on the thighs.
- Squat down with the elbows fully extended until the bar touches the floor or drop the bar to the platform if rubber

bumper plates are being used.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris, soleus, gastrocnemius, deltoids, trapezius

VERBAL CUES

- **Squeeze the bar from the floor**
- **Long arms during the pull**
- **Punch your hands to the sky**

COMMON ERRORS

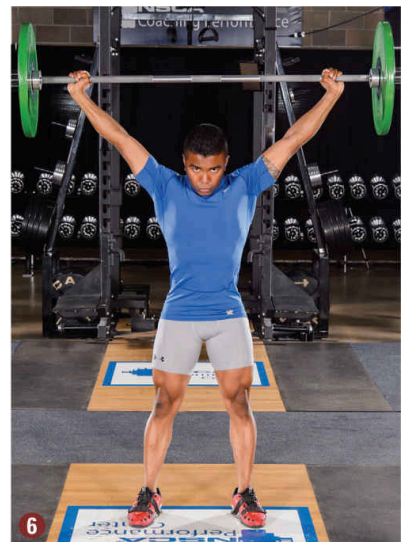
- **Letting the bar drift forward away from the shin**
- **Raising the hips too quickly as the bar is lifted from the floor**
- **Flexing the elbows prior to completing the triple extension**



End of second pull



Catch



End position

**44**

CLEAN GRIP MID-THIGH PULL (AND SNATCH GRIP VARIATION)

The performance of the mid-thigh pull with the clean or snatch grip requires the use of lifting blocks or a rack that allows the loaded barbell to be positioned so that the athlete initiates the movement from the power position (i.e., second pull position, knee angle ~ 125 - 145 degrees and hip angle ~ 140 - 150 degrees). When using the clean grip, the bar will generally be positioned across the upper thigh, depending on the athlete's individual anthropometrics (i.e., arm length). If the snatch grip is used, the bar will generally be higher on the thigh and in most instances sit across the hip crease as a result of the wider grip. To accommodate the wider grip used during the snatch grip mid-thigh pull, the loaded bar will need to be placed on higher blocks (or rack position) than what is used with the clean grip mid-thigh pull. While there are differences in the placement of the barbell between the clean and snatch grip variants of this lift the fundamental movement patterns associated with performing the mid-thigh pull are the same regardless of which grip is used.

Starting Position

The power position is the starting position for the clean grip mid-thigh pull (and snatch grip mid-thigh pull).

- Stand with the feet placed between hip- and shoulder width apart with the toes pointed slightly outward.
- Grasp the bar with a clean grip (i.e., slightly wider than shoulder-width apart) while using lifting straps with the elbows fully extended and pointing outward (i.e., arms internally rotated and the shoulder retracted). (If performing this exercise with a snatch grip, grasp the bar with a wider grip based on the grip established for the power snatch.)
- Move into a position where the trunk is in an upright position and there is a slight knee flex; the shoulder is above or slightly

behind the barbell and slack is removed from the body.

Note: When performing the mid-thigh pull the bar should not be placed at the mid-thigh position (i.e., midpoint between the iliac crest and the middle of the patella) because this will result in the athlete being in the wrong position (e.g., excessive torso lean). Due to the anthropometric differences (i.e., limb lengths) between athletes, the bar position will not be the same for all athletes. As such the coach should ensure that each athlete's individualized power position is used when performing this lift.

Upward Movement Phase

The upward movement phase mirrors the movements used when performing the second pull in the power clean (or power snatch if using a snatch grip).

- Rapidly extend the knees, hips, and ankles, (*Note:* it is important that the heels stay in contact with the floor for as long as possible in order to maximize force transference to the barbell.)
- Keep the bar as close to the body as possible.
- Keep the back neutral and the elbows pointing out to the sides.
- Keep the shoulders over the bar and the elbows extended as long as possible.
- When the lower body joints reach full extension, rapidly shrug the shoulders upwards with the elbows still fully extended and pointing to the sides.

Downward Movement Phase

During the downward movement phase, the barbell is returned to the blocks or rack and the starting position should be reestablished. It is important that the start position is reestablished prior to each repetition performed in a given set.

- In a controlled manner while maintaining an upright trunk position lower the barbell back to the starting position.
- While lowering the barbell it is important that the trunk remains rigid (i.e., in a neutral spine position).

- Avoid bouncing the barbell off the blocks.

Note: The mid-thigh pull can be performed with loads greater than the loads that are used when performing the power clean or power snatch. In fact, it is not uncommon for athletes to be able to perform the mid-thigh pulls with as much as 140% of the 1RM.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris, soleus, gastrocnemius, deltoids, trapezius

VERBAL CUES

- Drive the feet through the floor
- Stay on your heels as long as possible
- Shrug vertically with long arms

COMMON ERRORS

- **Not maintaining a vertical trunk position resulting in the hips being too high and the shoulder being placed in front of the bar. (*Note:* often a result of initiating the lift from the wrong position.)**
- **Performing the shrug motion before completing the triple extension of the lower body**
- **Flexing the elbows before completing the triple extension of the lower body**



Highest bar position (clean grip)



Highest bar position (snatch grip)

▶ 45 **CLEAN PULL FROM KNEE-BLOCK (AND SNATCH GRIP VARIATION)**

The clean pull from the knee-block (and snatch grip variation) is an exercise that is performed from a position that corresponds to the end of the first pull during the performance of the power clean (or power snatch). The start position corresponds to the end of the downward movement of the RDL exercise.

Starting Position

- Position the blocks so that the loaded barbell is at a height that corresponds to the athlete's patellar tendon.
- Stand with the feet placed between hip- to shoulder-width apart with the toes pointed slightly outward.
- Grasp the bar with a clean grip (i.e., slightly wider than shoulder-width apart) while using lifting straps with the elbows fully extended and pointing outward (i.e., arms internally rotated and the shoulder retracted). (If performing this exercise

with a snatch grip grasp the bar with a wider grip based upon the grip established for the power snatch.)

- Hold the back in a normal lordotic position while slightly flexing the knees and flexing at the hip so that the shoulder is in front of the bar.
- Ensure that the shins are perpendicular to the ground and the weight is centered between the feet.

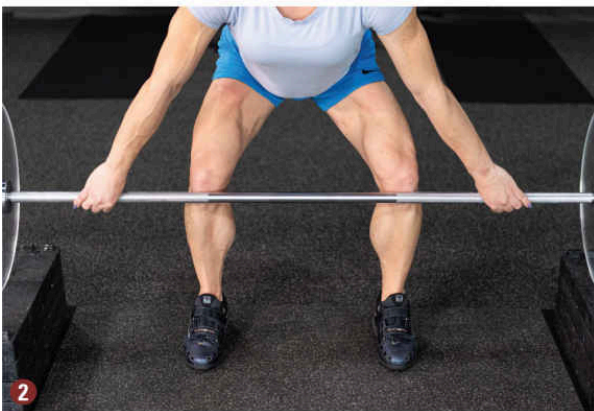
Upward Movement Phase: Transition

- As the bar rises just above the knees, extend the hips, while moving the knees forward and under the bar, to transition the athlete into the power position.
- Maintain bar contact with the thighs.
- Keep the back neutral or slightly arched and the elbows fully extended and pointing to the sides.

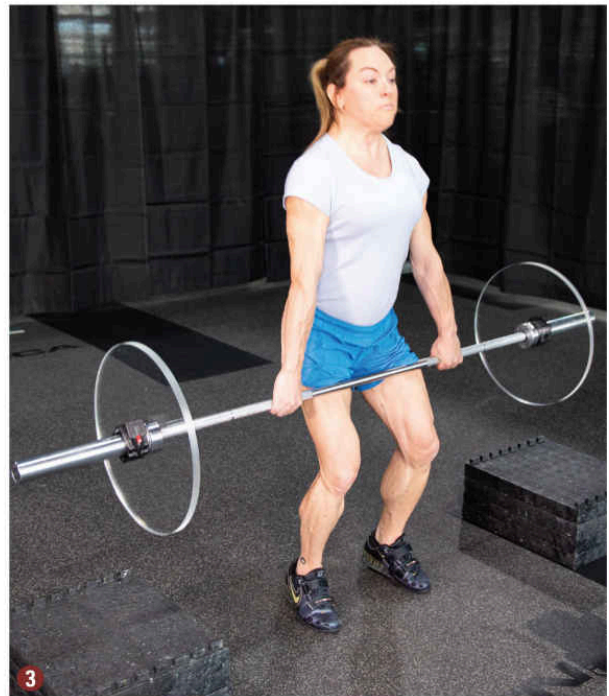
Note: The transition phase is similar to the RDL; in fact, weightlifters use the RDL to strengthen this movement pattern.



Starting position (clean grip)



Starting position (snatch grip)



Upward movement phase: Transition

Upward Movement Phase: Second Pull

- Rapidly extend the hips, knees, and ankles, culminating with a shrug. (*Note:* It is important that the heels stay in contact with the floor for as long as possible in order to maximize force transference to the barbell.)
- Keep the bar as close to the body as possible.
- Keep the back neutral and the elbows pointing out to the sides.
- Keep the shoulders over the bar and the elbows extended as long as possible.

- When the lower body joints reach full extension, rapidly shrug the shoulders upward with the elbows still fully extended and pointing to the sides.
- As the shoulders reach their highest elevation, flex the elbows to begin pulling the body under the bar.
- Due to the explosive nature of this phase, the torso is erect or slightly hyperextended, the head may tilt slightly back, and the heels may lose contact with the floor.

Downward Movement Phase

During the downward movement phase, the barbell is returned to the blocks or rack and the starting position should be reestablished. It is important that the start position is reestablished before each repetition performed in a given set.

- Flex at the knee and hip while lowering the barbell to the blocks.
- Maintain a neutral back and ensure the elbows are pointing to the sides.
- Avoid bouncing the barbell off the blocks.

Note: The mid-thigh pull can be performed with loads greater than the loads that are used when performing the power clean or power snatch. In fact, it is not uncommon for athletes to be able to perform the mid-thigh pulls with as much as 140% of the 1RM.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris, soleus, gastrocnemius, deltoids, trapezius

VERBAL CUES

- **Hips high and shoulder in front of the bar**
- **Tight back**
- **Shrug vertically with long arms**

COMMON ERRORS

- **Dropping the hips and moving the shoulder behind the bar**
- **Coming to the toes too soon**
- **Rounding the back and not maintaining a normal lordotic position**



Upward movement phase: Second pull



Upward movement phase: Catch

KEY TERMS

alternated grip
clean grip
closed grip
false grip
five-point body contact position
forced repetitions
free weight exercises
grip width

hook grip
liftoff
neutral grip
neutral spine
out-of-the-rack exercises
overhand grip
over-the-face barbell exercises
partner-assisted reps
power exercises
pronated grip
range of motion (ROM)
snatch grip
spotter
sticking point
structural exercises
supinated grip
supine
underhand grip
Valsalva maneuver

STUDY QUESTIONS

1. Which of the following grips is used for the lying triceps extension exercise?
 - a. open, pronated
 - b. open, alternated
 - c. closed, pronated
 - d. closed, alternated
2. An athlete performing which of the following workouts has the greatest need for a weight belt?
 - a. deadlift at 90% of the 1RM
 - b. deadlift at 50% of the 1RM
 - c. bench press at 95% of the 1RM
 - d. power snatch at 70% of the 1RM

3. All of the following require one or more spotters except:
 - a. power clean
 - b. shoulder press
 - c. bench press
 - d. back squat
4. When an athlete is performing the barbell bench press exercise, all of the following describe the actions of the spotter except:
 - a. Stand erect behind the bench with the feet shoulder-width apart.
 - b. Grasp the bar with a closed, alternated grip inside the athlete's hands.
 - c. Tell the athlete when the bar will be lifted off the supports.
 - d. Be sure the athlete has complete control of the bar at the conclusion of the liftoff.
5. Which of the following describes the proper breathing technique for the shoulder press exercise?
 - a. Exhale through the sticking point on the upward movement phase.
 - b. Inhale before the sticking point on the upward movement phase.
 - c. Exhale before the sticking point on the downward movement phase.
 - d. Inhale through the sticking point on the downward movement phase.